MUSKAN SUMAN

AI / ML Engineer - Computer Vision & LLM Systems

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Professional Summary

AI/ML Engineer who ships from training custom neural networks to deploying scalable inference systems on AWS. Passionate about pushing intelligence to the edge, where latency is real and failure isn't an option. Built production-grade ADAS pipelines processing real-time safety events for commercial fleets and LLM-powered assistants using RAG, multi-agent architecture, and vector search. M.Tech from IIT Jodhpur. Not just a researcher an engineer who owns the full stack, from model to metal.

Core Skills

Languages	Python, C++, SQL
AI / ML	PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, YOLO, Transformers, ONNX
GenAI & LLMs	RAG Pipelines, Prompt Engineering, VLMs, VLLM, pgvector, NLP, Embedding Models
Data	NumPy, Pandas, Feature Engineering, Data Augmentation, Synthetic Data Generation
MLOps & Backend	FastAPI, Docker, MLflow, Git, CI/CD
AWS Cloud	Bedrock, SageMaker, EC2, S3, Lambda, DynamoDB, RDS, Aurora PostgreSQL, ECR , AWS Beanstalk

Experience

AI / ML Engineer — Novus Hi-Tech

Jul 2024 – Present

Gurugram, Haryana

- **Fleet-GPT — AI-Driven Fleet Management Assistant**
  * Designed and integrated a Retrieval-Augmented Generation (RAG) system using LLMs, AWS Bedrock, and **pgvector** on Aurora PostgreSQL to enable real-time querying of structured and unstructured fleet operational data.
  * Developed scalable backend pipelines for workflow orchestration, dynamic tool calling, semantic retrieval, and multi-agent integration while optimizing prompt and retrieval strategies for improved response accuracy and low-latency inference.
- **Driver-Monitoring False-Alert Reduction Pipeline**
  * Engineered a production-scale **ADAS driver-monitoring AI pipeline** using **YOLOv15, YOLOv11-Pose, OpenCV, FastAPI, AWS Lambda, DynamoDB, and EC2** to re-validate safety events including phone use, drowsiness, smoking, FCW, and lane-departure warnings for commercial fleet vehicles.
  * Developed a multi-stage ML inference system combining **object detection, pose estimation, geometric reasoning, and temporal filtering** to suppress false-positive alerts before downstream fleet-manager escalation, deployed through an asynchronous queue-backed FastAPI service for real-time inference.
- **Drowsy Disambiguator System — Driver Drowsiness Detection Pipeline**
  * Designed and trained **DBDNet**, a lightweight **1D-CNN with Multi-Head Self-Attention (~13K parameters)** for real-time driver drowsiness detection, achieving **86.7% accuracy, 0.90 F1-score, and 0.99 ROC-AUC** on validation data.
  * Built and deployed an end-to-end driver monitoring pipeline using **BlazeFace, TFLite eye-state classification, blink analytics, FastAPI, Docker, and AWS ECR**, improving **heavy-drowsy retention from 33% → 100%** and **moderate-drowsy retention from 54.5% → 91%** over the previous rule-based baseline.
- **Synthetic Data Generation & Domain Adaptation**
  * Developed a synthetic data generation and augmentation pipeline using diffusion-based image synthesis and prompt engineering for computer vision training tasks.
  * Performed representation-space and semantic-domain analysis using CLIP embeddings, PCA visualization, and TF-IDF techniques to study synthetic-to-real distribution shifts and improve downstream detection performance.

Education

Indian Institute of Technology (IIT), Jodhpur
Jul 2023 – May 2025

M.Tech — Robotics and Mobility Systems — CGPA: 7.89 / 10
Jodhpur, India

Thesis: Vibration Estimation in Autonomous Vehicles Using IMU Data

* Applied EMD, VMD, and Hilbert Transform for vibration-frequency analysis using IMU sensor data.

* Developed parametric models for road-induced vibration estimation supporting suspension calibration and sensor-fusion systems.

Nitte Meenakshi Institute of Technology
Jul 2018 – May 2022

B.Tech — Aeronautical Engineering — CGPA: 8.55 / 10
Bangalore, Karnataka

Key Projects

AutoValuator AI — LLM-Powered Car Price Prediction Assistant GitHub | Live Demo

- Developed an end-to-end conversational vehicle valuation platform integrating a **Random Forest regression model** with **Llama-3.3-70B (Groq)** for natural-language feature extraction, explainable price prediction, and multi-turn interactions using **Redis-backed memory**.
- Engineered and deployed a production-grade **FastAPI** microservice with **JWT/API-key authentication, rate limiting, Prometheus/Grafana monitoring, and Dockerized infrastructure**, enabling scalable deployment on **Render** and **Streamlit Cloud**.

Terrain Traversability Segmentation for Autonomous Ground Robots GitHub

- Developed semantic and instance segmentation pipelines for autonomous driving scenes using **UNet**, **YOLOv8**, and **Detectron2** on a custom road-scene dataset comprising **1,000+ annotated images**.
- Implemented **UNet from scratch** with IoU loss and advanced data augmentation techniques, while achieving an **mAP@50 of 0.91** using **YOLOv8** for instance segmentation.

Hybrid Physics-ML System for Wind Farm Layout Optimization GitHub

- Developed a hybrid optimization framework integrating a physics-based **Jensen wake model** with an **XGBoost surrogate model (R^2=0.90)** to accelerate wind farm layout evaluation and Annual Energy Production (AEP) prediction.
- Applied constrained optimization techniques including **SLSQP, L-BFGS-B, and multistart optimization**, where the ML-guided SLSQP approach achieved a **20.45% improvement in Annual Energy Production** over a random baseline.